Grounding Without Corrective Control: Truth-Tracking Profiles for Large Language Models

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Abstract

Recent work makes a strong case that at least some large language model representations have content or reference. That leaves a different question open. Grounding can secure content, reference, or correctness conditions without supplying the live, sufficiently independent routes through which a discrepancy with the target is detected and made to alter production, uptake, withdrawal, or later behaviour. Call the weaker relation answerability and the stronger one corrective control. This paper analyses both as a profile: a pattern of participation in routes that give target access, integrate representations, handle discrepancies, and register practical success or failure. Answerability comes in degrees; corrective control is the case where answerability includes both detection and an effective route to repair.

Language models provide the pressure case, with the text-only profile taken as a limiting position rather than a description of a product. Text-trained systems inherit dense textual traces of testimony and local coherence, which leaves them derivatively answerable to corrections already made in the source practices, and participate weakly in perception, measurement, intervention, and practical correction, which leaves them without corrective control wherever the relevant target isn't reachable through a live checking route. The result is a familiar failure signature: fluent, coherent outputs that hold up until a task needs an independent route to the facts. Self-consistency, retrieval, tools, code execution, multimodal input, and preference or action feedback should help where they supply the route a task requires and not otherwise, so the prediction is an interaction between intervention and task rather than a general gain. A crossed design, varying route availability against task requirement, can test it. Surface improvement and truth-tracking improvement can come apart.

Keywords: large language models; grounding; answerability; corrective control; projectibility; truth-tracking; philosophy of AI

AI USE. The large language models Anthropic Claude Opus 4.8, Fable 5, and Opus 5; and OpenAI Codex using GPT-5.5 served as drafting and editing aids throughout the preparation of this paper. I am responsible for all theoretical claims, arguments, errors, and interpretive choices.

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1 Introduction: from grounding to corrective control

Do the representations inside large language models (LLMs) connect to the world? The question is often put as a yes/no. Bender and Koller (2020) say no: a system trained on form alone, like an octopus eavesdropping on telegraph traffic, learns the statistics of language apart from what language is about. Others reply that such systems meet at least some conditions on reference, or that sensory grounding can be optional for thought in the first place (Chalmers, 2023; Coelho Mollo & Millière, 2026).

But that opposition only starts the inquiry. The debate descends from the symbol-grounding problem (Harnad, 1990): how can symbols, or symbol-like states, get meaning independent of the interpretations supplied by users? For LLMs, recent work has already pluralized the issue. Pavlick treats grounding as a family of questions about symbols and use; Coelho Mollo and Millière distinguish kinds of grounding; Chalmers separates sensory grounding from thought (Chalmers, 2023; Coelho Mollo & Millière, 2026; Pavlick, 2023).

The recent literature has been closing the sceptical routes one at a time, and closing them on the side of content. Reference can be secured by the natural histories of a model's inputs (Mandelkern & Linzen, 2024); intentions are the wrong place to look (Pepp, 2025); the argument from communicative intention fails at both premises (Attah, 2025); linguistic metasemantics delivers meaning where mental metasemantics doesn't (Grindrod, 2024); and meaning can be grounded in a corpus rather than in the world (Havlík, 2024). Ruyant reaches the further conclusion that what such systems represent is a class of appropriate linguistic production rather than the world at all (Ruyant, 2026).

I contest none of it. No party to that literature is my opponent, and nothing below turns on any of them having made a mistake. Some of this work does discuss misrepresentation, error, and repair.[1] What none of it supplies is an account of the live, task-indexed routes through which a fresh discrepancy becomes detectable and revision-producing. Establishing content and establishing corrective control are different achievements, and the second doesn't follow from the first.

Pluralizing grounding still leaves open which dimensions explain truth-relevant success and failure across systems. Truth-tracking success depends on distinct STABILIZERS that ordinarily operate together: perception, testimony, coherence, measurement, intervention, error-correction, and practical success.[2] A system can take part in some of these and sit out others. Text-only and weakly tooled language models make that possibility especially visible.

I argue that these stabilizers form a PROFILE, and that a system's profile warrants bounded projections about the successes picked out by the predicate *true*. A profile, in this sense, isn't a checklist of individually necessary conditions. It's a pattern of routes whose presence, absence, and coupling let us project how a system will do in nearby cases (Goodman, 1955; Khalidi, 2013).

LLMs matter because they separate routes that human epistemic life normally bundles together.

[1] Coelho Mollo and Millière build the possibility of misrepresentation into their account, and Attah (2025) discusses conversational repair. Neither supplies an account of the system-level routes at issue here, which is the point, and neither is committed to denying that such routes matter.

[2] The term isn't Millikan's. On Grindrod (2024)'s reading of her teleosemantics, a STABILIZING FUNCTION is conventional and inter-agent: it sits between speaker and hearer and explains why a linguistic device keeps being reproduced, and he puts that notion to work on language models. A stabilizer here is epistemic and world-directed: it explains why a representation is answerable to how things are. The distinction matters for Section 7, since a device can have a robust stabilizing function in Millikan's sense while participating only derivatively in the stabilizers in mine, which is arguably the text-only case.

The claim is clearest for systems whose contact with the world remains mediated mainly by human text. Retrieval, tools, code execution, multimodal input, and action-guided feedback change that inheritance-heavy profile. The question is which route they add and where improvement should transfer.

Comparison drives the argument. Self-consistency should help most where the missing support is coherence. Retrieval should help most where the missing support is record access. Preference feedback should improve conversational fit before it improves answerability, the relation that lets discrepancies bear on what a system produces. Multimodal input should help some perceptual tasks before it helps calibrated measurement. Success for the profile view depends on these contrasts illuminating LLM behaviour better than a single verdict about grounding.

A modest lesson follows, and it's about epistemic architecture rather than about truth. Correspondence, coherence, pragmatism, reliabilism, deflationism, and social epistemology each foreground a relation or process that can contribute to answerability. They're not thereby competing theories of one stabilizer, and the profile isn't a reconciliation of them. I aim to make the architecture precise enough to guide comparison.

Three things are new here. The first is the isolation. The live, task-indexed routes by which a fresh discrepancy reaches production are picked out as an explanandum in their own right rather than as a corollary of content. The claim isn't that anyone took reference to entail error-correction. It's that a positive grounding verdict settles what a system's representations can mean or refer to while licensing no inference about which fresh errors a deployed arrangement can catch, and what follows marks where that inference stops. The second is an account of what closes the gap, stated as five graded features of the output-producing arrangement rather than as a list of epistemic goods, with answerability graded beneath it and corrective capacity kept apart from how effectively it gets realized. The third is a prediction the account can lose: because interventions supply particular routes, their benefit should depend on which route a task requires, so the expected result is an interaction between intervention and task rather than a general gain, and indiscriminate transfer would count against the carving.

Section 2 first fixes the target: the relevant cases are representational successes (beliefs, assertions, inscriptions, measurements, models, inferential outputs), with truth as an abstract object left to one side. Section 3 sets out the stabilizers, their Boydian lineage, and the four positions they occupy. Section 4 says how the profile supports projection under counterfactual perturbation, and separates stability from corrective control. Section 5 shows how calibration history handles conflict and transfer. Section 6 takes objections. Section 7 fixes the bearer, works through the interventions and what each adds, and closes on the transfer prediction. Section 8 concludes.

2 The target: truth-relevant representational success

Before the profile can be applied, the target has to be fixed. Truth, considered as an abstract property or as the role of the truth predicate, lies outside the class of things with perceptual, testimonial, or instrumental mechanisms as parts. The target cases are representational successes: beliefs, assertions, inscriptions, measurements, models, and inferential outputs assessed as getting things right.

Some count as central TRUTHBEARERS. Beliefs and assertions can be true or false in the ordinary sense. Others are neighbouring cases that support truth-ascription by yielding, recording, or constraining claims: a measurement, a map, a statistical model, or an experimental display may be accurate, calibrated, or well-fitted in ways that make some associated propositions true. The profile concerns that broader space of truth-relevant representational success.

Truth-relevant work often gets distributed. A thermometer reading, a graph, or a database entry can make true assertion possible before anyone forms the final assertion. A map can preserve spatial relations well enough to support successful claims about where things are. A model can constrain which inferences remain live. The account is about the background that lets such outputs support getting things right.

In a lead-exposure case, the assay reading, the calibration log, the database entry, and the clinician's assertion do different representational jobs. Only the last may be the explicit assertion that the sample contains 3.2 mg of lead, but the others make that assertion available, checkable, and defeasible.

So the paper leaves a reductive definition of truth to one side. Deflationary and minimalist theories can insist that the truth predicate has a thin logical or generalizing role, with a substantive causal story kept out of the basic theory of truth (Horwich, 1998). That point stands. The profile account asks a different question, namely why some truth-relevant representational successes travel together, support projection, and fail in patterned ways.

Correspondence can also remain in place as one stabilizer within the larger account. Correspondence theories preserve the idea that truth involves answerability to how things are (David, 1994; Kirkham, 1992).

Here, ANSWERABILITY names a constraint relation. A representation is answerable to a target when discrepancies between the representation and the target can bear on the production, uptake, correction, or withdrawal of the representation. Strictly, profiles belong to arrangements, and an output is answerable relative to the arrangement that produces, maintains, and revises it; Section 7 fixes that bearer. The shorter locution is used where no confusion follows.[3] A calibrated instrument has answerability because relevant discrepancies can show up in checks, recalibrations, and downstream failures. A coherent but disconnected generator has weaker answerability where the relevant fact is disconnected from the production route. Profiling these routes shows how they combine and come apart.

A fact can bear on an output through delayed routes: later uptake, audit, replication, practical failure, or correction of future behaviour. The answerability is stronger when the route is specific, repeatable, and close enough to make the relevant error visible. It's weaker when the route is delayed, generic, or only loosely tied to the claim.

Scale drift gives the contrast practical as well as semantic force. Repeated weighing, standard weights, and failed downstream uses can expose the drift. If a witness is mistaken, records, other witnesses, and later events can defeat the report. If a model generates a plausible citation from textual pattern alone, the output has so far been shaped by genre, with those checks absent from the production route. It may still be right, and its rightness is less supported by the route that produced it.

A single sentence can differ in answerability. "The sample contains 3.2 mg of lead" is strongly answerable when it's generated by a calibrated assay whose logs, standards, and downstream checks

[3]The lexical history of *true* is suggestive evidence only. The *Oxford English Dictionary* records Old English senses of loyalty, trustworthiness, and veracity, early senses of accordance with fact, and later medieval senses of accurate fit, proper purpose, and mechanical alignment (Oxford English Dictionary, 2026). A ruler, balance, wheel, or surface can be true when it meets the standards of the practice in which it's checked. That history doesn't settle the analysis, but it fits the constraint relation used here.

can expose error. The same string is weakly answerable when a text-only model produces it because such numbers fit the genre. The truth conditions haven't changed. The route by which the claim became available has.

Pragmatist theories mark a different stabilizer. Inquiry, action, and successful practice can correct locally coherent or socially inherited representations with weak world-connection (Dewey, 1941; James, 1907). That doesn't make truth identical with what works. Practical success helps representational systems become answerable to what they're about.

I use the account to explain the stabilizing background that makes truth-ascription, accuracy-ascription, and truth-relevant representational success projectible across cases. We ask why some outputs fall inside the extension of truth-relevant success, why nearby outputs fall outside it, and which features let us project from a partial description of a case to its likely epistemic behaviour. Goodman's projectibility problem enters here: the task shifts from attaching the word *true* case by case to finding which patterns make truth-relevant classifications reliable guides to further expectation.

Read "truth-tracking" in that modest sense. It names patterned support for representational success, while Nozick's truth-tracking account targets subjunctive conditions on knowledge (Nozick, 1981). Patterned support of this kind lets us expect some representational successes to survive checks, perturbations, and neighbouring applications better than others.

Boyd's accommodationism supplies the background for that move. Reference and truth depend on more than definitions in isolation; they're achievements of practices that align perceptual, instrumental, cognitive, and representational activity with relevant causal structures (Boyd, 2021). That frame can be accepted while asking a narrower question: how strongly does a given system participate in the practices and mechanisms that make such alignment reliable?

Language models bypass some of these stabilizers and are assessed through others, which is what makes the lesson plain. A text-only model may inherit textual traces of testimony, statistical regularities, and local coherence pressures with attenuated perception, measurement, intervention, and ordinary practical correction. That profile is enough to make truth-relevant assessment possible, uneven, and empirically diagnostic.

3 Truth-tracking as a stabilizer profile

Goodman's new riddle fixes the projectibility problem: which predicates support projection from known to unknown cases (Goodman, 1955)? Here, that question concerns truth-relevant representational successes: which stabilizer patterns let us project from partial evidence to likely success or failure?

Khalidi's account supplies the network discipline behind this profile talk. Natural categories earn their status by supporting prediction and explanation, and they do so when associated properties stand in causal relations that make some properties projectible from others (Khalidi, 2013).

I adapt the framework narrowly to the representational successes fixed in Section 2; the stabilizers explain why those cases support projection. Perception, testimony, coherence, measurement, intervention, error-correction, and practical success are the routes through which representational outputs become more or less answerable to the world. Khalidi's later "nodes in causal networks" formulation supplies the graph vocabulary. On his account a kind is a node, a highly connected vertex whose core properties give rise to derivative ones, and the network is what the node sits in (Khalidi, 2018). A profile isn't a kind in that sense. What the account takes from him is the discipline that connectivity does the projective work, not the claim that truth-tracking is a node with a core.

Listed flat, they look like a miscellany of epistemic goods, and they aren't coordinate. They occupy four positions in an architecture. TARGET ACCESS covers the routes by which the world reaches a representation: perception, testimony and records, measurement, and intervention. INTEGRATION covers relations among representations, chiefly coherence and inference. DISCREPANCY HANDLING covers what happens once a mismatch appears: detection, attribution, correction, withdrawal, updating. PRACTICAL FEEDBACK covers success and failure in use.

An item can occupy more than one position, which is why the flat list misleads. Measurement supplies target access, generates discrepancies, and takes part in calibration. Practical failure can reveal that perception, testimony, or inference has drifted. Table 1 sets the positions out.

Table 1: Positions in the answerability architecture.

Position	Routes	What its absence costs
Target access	perception; testimony and records; measurement; intervention	the world can't reach the representation at all
Integration	coherence; inference	inputs can't be brought into contact with one another
Discrepancy handling	detection; attribution; correction; withdrawal; updating	a mismatch can arise and leave nothing changed
Practical feedback	success and failure in use	error stays costless, so nothing selects against it

Section 4 grades discrepancy handling in particular. Target access, integration, and practical feedback say where constraint comes from; discrepancy handling says whether constraint does anything when it conflicts with what the arrangement produced. That's why error-correction sits oddly in a flat list: it isn't a further source alongside perception and testimony, but the dimension on which an arrangement with plenty of sources can still fail.

Calling the result a profile marks two points. First, different routes can dominate in different domains. Proof norms matter more in mathematics than ordinary perception does; calibration matters more in laboratory measurement than conversational fluency does. Second, the epistemic force often lies in the relations among routes. Testimony matters differently when it's backed by records, instruments, and correction than when it's an isolated report.

Boyd supplies the lineage for this profile view. Homeostatic property cluster theory explains how real kinds can be projectible in the absence of essences (Boyd, 1991, 1999). His accommodationism, introduced in Section 2, adds the detail that matters here. Perceptual skills, instruments, inferential routines, social arrangements, and representational uses each contribute when they help a practice align with relevant causal structures (Boyd, 2021), which is what makes the routes below a list of contributors rather than a list of epistemic goods.

Accommodation, in this sense, is the gradual fitting of representational practice to the structures it's used to track. A scientific term, an instrument reading, or an ordinary object name becomes more reliable as perception, measurement, inference, correction, and use are adjusted together. Definitions alone don't secure the fit; practices that make error detectable and revision possible maintain it.

Consider *fever*. Feeling hot, thermometer readings, calibration standards, clinical thresholds, records, and revised estimates of normal body temperature have adjusted one another over time. The term stays useful because the practice has routes for detecting error and changing how the target is measured.

Within that lineage, homeostasis marks the important special case. Links are actively maintained: mechanisms correct one another, and the properties co-occur far more reliably than chance. The truth-tracking profile has tightly coupled regions, and the connections thin toward its edges. It can project without the whole profile being literally homeostatic.

Keeping that distinction in view limits the claim, since a set of features can support projection without corrective feedback actively maintaining it (Reynolds, 2026a). Section 4 makes the gradations precise.

I narrow Boyd's question to text-trained participants. An LLM can inherit stabilized linguistic and testimonial patterns while its instrumental, perceptual, and practical routes for answerability remain attenuated.

At the explanatory level, the account stays continuous with Boyd; at the system level, it stays diagnostic. Accommodationism tells us why representational practices can be answerable to causal structure; the profile account tracks how strongly a given system participates in the stabilizers that make that answerability possible. LLMs serve as the pressure case because they inherit accommodation through human text and weaken several world-corrective paths.

Two jobs follow. First, the picture identifies the tightly coupled core. In everyday perception and basic coordination, perception, action, memory, testimony, and practical failure – the routes where error has been costly longest – have repeatedly corrected one another. That coupling explains the familiar language-model failure signature: remove the perceptual and verificational nodes, and representational success should fail in a specific way (Section 7).

Second, it places the loose cases. Toward the edges, in frontier science and remote historical fact, the nodes are real but sparsely linked. Error cost is delayed, diffuse, or beyond current instruments. There may be inference norms, expert testimony, and coherence constraints, but fewer shared routes of correction. Section 5 makes that diagnosis explicit: shared selection or training history, accumulated correction procedures, and measurement infrastructure are often thin at the edges.

Reading the profile as a network still leaves boundaries to be fixed. Onishi and Serpico (2022) show that carving a kind from a causal graph stays relative to the chosen level of abstraction; Craver (2009) makes the same point for mechanism talk.

So the paper fixes the epistemic purpose. Following Khalidi's distinction between epistemic purposes and non-epistemic practical or aesthetic ones, the profile is individuated by representing a target domain well enough for relevant inferential, practical, or corrective success under perturbation (Khalidi, 2013). Relative to that purpose, its connections can thin at the edges and the profile can still hold.

4 Why the profile is projectible

Section 2 fixed the target cases, and Section 3 identified the stabilizers. Projectibility concerns the relation between them. The question is whether a partial description of an output and its stabilizing background lets us predict further truth-relevant success or failure.

Perturbation is what makes the purpose just fixed do any work. It distinguishes lucky fit from robust answerability. A representation may succeed in the training cases, the familiar environment, or the current conversational context. It projects more strongly when it continues to guide action, inference, measurement, and correction after conditions change.

The causal backbone stays deliberately light. In Pearl's hierarchy, the profile begins as association among stabilizers, mechanism is identified by perturbation, and projectibility is the counterfactual question of what would hold nearby; Pearl's intervention operator formalizes that perturbation test (Pearl, 2009, 2010). Woodward supplies the manipulability reading: a stabilizer is explanatory when interventions on it change the cluster in stable, predictable ways (Woodward, 2001, 2003). This argument doesn't assume a specifiable graph; it uses the ascent from association to intervention to counterfactual projection rather than a full structural causal model.

That machinery establishes invariance, which is the weaker of the two things at issue here. World-directed claims form a partial order (Reynolds, 2026b). Stability says a profile recurs or survives declared variation without saying what produces it. Directed dependence says a specified non-accidental relation makes one feature informative about another. Neither entails the other, which is what makes the order partial rather than a ladder, and maintenance includes both. Corrective control, defined next, sits above maintenance.

CORRECTIVE CONTROL

↑

MAINTENANCE

↗ ↖

STABILITY DIRECTED DEPENDENCE

Upward edges mark inclusion of the lower claim's requirements. They don't rank arrangements, and the ordering over the five features below is a separate matter that leaves many pairs incomparable.

Corrective control adds detection and effective correction, and it's target- and range-relative rather than a property an arrangement has outright. Its strength, for a given target and perturbation range, depends on five features, named here so later sections can refer to them without counting. Target access means target-relevant variation reaches the arrangement through an operative route. Detectability means a discrepancy can be registered by that arrangement against a standard it can consult. Checking-route independence means the checking signal doesn't merely reproduce the production route's error. Attribution means the discrepancy is specific enough to guide a response. Effective uptake means that when a relevant discrepancy occurs, is detected, and is attributable, the arrangement can make a change that reduces it within the declared horizon.

These are dimensions, not switches, and nothing turns on their being independent. Each admits of degree: a checker's errors can be partly correlated with the generator's, attribution can be more or less precise, uptake can be delayed or probabilistic. They also run together in practice, which Section 3 should lead one to expect, since a single route can occupy several positions. Giving a checker access to repository state adds target access and may decorrelate its errors at the same time. An arrangement is under stronger corrective control as more of these hold more fully, and the interesting comparisons are between arrangements that differ on one of them. That ordering is partial. Two arrangements trading decorrelation against attributability aren't ranked by this account, and nothing here licenses a weighted score across the five.

All five are capacities, and stating effective uptake as one takes some care. The tempting alternative is a rate, on which a route contributes in proportion to how often detection alters production. That collapses too much. How often an arrangement corrects depends on how often relevant discrepancies arise, so a near-flawless arrangement in a benign environment would score low while a poor generator triggering a mediocre checker would score high. Counting any alteration also counts alterations in the wrong direction.

So the fifth takes the conditional and directional form above. Given a discrepancy that occurs, is detected, and is attributable, can the arrangement make a change that reduces it? That's a question about architecture. What varies separately is REALIZED CORRECTIVE EFFECTIVENESS: conditional on those same antecedents, how often the arrangement does make an appropriate target-improving change, and how often it makes an unnecessary or harmful one. Observed repair rates are evidence about capacity and effectiveness. They aren't what corrective control consists in.

Checking-route independence cuts across the other four rather than sitting beside them. It's a property of where a checker came from, what it can see, what it's optimizing, and how its errors covary with the generator's, and any of the first, second, or fourth features can be present with it or without it.

These conditions fix the relation to answerability rather than competing with it. Answerability comes in degrees, and its strength depends on how specific, repeatable, and proximate the route is. Corrective control is the stronger case, where answerability includes both discrepancy detection and an effective route to repair. So an arrangement can be answerable in some measure while lacking corrective control over the targets at issue, which is the situation Section 7 describes.

A thermometer makes the gap concrete, though not the uncalibrated one. Standards existing somewhere don't make a discrepancy detectable by an arrangement that never consults them, so an uncalibrated thermometer already fails detectability and the example doesn't isolate what it should. Take instead a thermometer wired to a calibration monitor that flags drift visibly, in a shop where every alert is ignored. Target access, detectability, independence, and attribution are all present, since the alert is raised, is correct, and identifies the fault. What's missing is any route from the alert to what the arrangement does, and because the people ignoring it are part of the arrangement, that's a missing capacity rather than a low score.

An approval step that users clear 97 percent of the time is a different case. There the route exists and is sometimes taken, so effective uptake is present and realized effectiveness is low. Telling those two apart is what a rate formulation can't do.

Different regions of the profile are claimed at different rungs. Everyday perception and basic coordination carry routes that detect departures and repair them, so corrective control is claimed there and evidenced by the repairs. Frontier science and remote historical fact often support stability and directed dependence without it. The language-model case in Section 7 turns on that gap: text-only systems inherit stable and densely dependent textual structure while participating weakly in detection and repair.

A simple example helps. A weather claim repeated from an old webpage may be true when the weather is stable and the page is recent. It projects weakly once the date, location, or local conditions change. A claim constrained by live measurement, calibrated instruments, and error-reporting practices projects more strongly because nearby changes have routes through which they can affect the representation.

Survival and reproduction mark limiting biological cases of this structure. Hoffman-style interface arguments warn that adaptive success and veridical representation can come apart (Hoffman et al., 2015). Martínez (2019) argues that usefulness can drive representations toward truth under specified conditions. The lesson for this paper is methodological: usefulness becomes truth-relevant when the task, environment, and correction channels make error costly in ways that expose the relevant mismatch.

Outside biology, institutions and technologies often supply the stabilizers. In science, law, medicine, engineering, testimony, and model evaluation, the relevant routes include replications, instruments, standards, records, peer challenge, and downstream practical failure. These routes help a truth-relevant output become robust under perturbation.

Useful classifications show why "works in practice" is too coarse on its own. A classification can succeed because it tracks a real structure, because the environment has been arranged around it, or because a task rewards a proxy. Only structure-tracking, and some forms of environmental arrangement, support the kind of projection at issue here. The profile has to say which route is doing the stabilizing.

Coupled routes support projection. A perceptual judgement supported by immediate appearance alone projects weakly if it's easily overturned by lighting, perspective, testimony, or measurement. The same judgement projects more strongly when it's embedded in a practice where perception, action, memory, testimony, and correction have repeatedly calibrated one another.

Projection comes in degrees. The question is how much of the relevant correction background the case carries with it. The more the routes have been mutually calibrated, the more confidently success in one nearby case licenses expectations about another.

Across domains, different stabilizers can take priority. Some mathematical and logical cases project through inferential structure with little perceptual input. Some scientific cases project through instrumentation and replication instead of ordinary observation. Some testimonial cases project because the institution has mature roles, records, and correction channels. The projectible unit is the coupled pattern.

Edge cases follow from the same structure. Where stabilizers thin or fail to calibrate one another, truth-relevant classifications should project less strongly. Frontier science and remote history should be treated as thinner regions of the profile. They may still involve real stabilizers, but the pattern supports weaker expectations about agreement, correction, and future success.

Applied to language models, the diagnosis becomes concrete. A system that strongly inherits textual traces of testimony and local coherence, with weak participation in perception, measurement, intervention, and practical correction, should show a specific failure profile. It can produce fluent outputs that look truth-relevant from inside the textual profile and fail under perturbations that require independent contact with the world.

Tool use, retrieval, multimodal input, expert review, and action-guided feedback matter because they change the stabilizer pattern. They may strengthen projectibility in some domains, leave it weak in others, or introduce new routes to error. The profile view predicts comparative differences among systems and tasks.

5 What profile structure adds to component theories

Profile structure contributes when it answers questions that are hard to formulate if one stabilizer is treated as the main unit of assessment. The answer turns on coupling: the way mechanisms have calibrated one another can carry the evidential fact.

Component theories enter as local successes. Reliabilism foregrounds reliable processes (Goldman, 1979). Coherentism foregrounds integration, with input needed for truth-conduciveness (BonJour, 1985). Pragmatist accounts stress correction through successful practice (Dewey, 1941; James, 1907). Correspondence accounts preserve answerability to the world (David, 1994; Kirkham, 1992). Social epistemology treats testimony and institutions as genuine epistemic sources (Coady, 1992; Goldberg, 2010; Lackey, 2008).

Taken together, these claims form part of the positive view. Reliable production, coherence, answerability, action, testimony, and institutions each contribute to stabilization. The question is what happens when those stabilizers are separated, recombined, or moved into a new system.

After those successes are granted, the profile view begins with the pattern of calibration among stabilizers: perception checked by testimony, testimony disciplined by memory and records, coherence constrained by measurement, measurement corrected by intervention, and practice pressured by failure. Read as a causal network (Section 3), the profile has structure that source-by-source assessment leaves in the background: the strength and history of the links between nodes.

A source's evidential force can change when the rest of the correction network changes. A locally reliable route may stop projecting when it's removed from the practices that made it reliable.

LLMs sharpen this point for reliabilism in particular. The relevant process is rarely just "the model generated this text". It may include pretraining data, filtering, instruction tuning, retrieval, tool calls, prompting, deployment constraints, post-hoc checking, and institutional uptake. It may also include reinforcement learning from human feedback (RLHF). Reliability often attaches to that system-profile. A cleanly bounded internal generator may be too narrow. Reliabilism can preserve its central insight by individuating the process at the same level at which the stabilizers actually operate.

Reliability and answerability cross-cut. A route can be reliable while weakly answerable, and answerable before it's reliable. It may be reliable in a familiar environment but weakly answerable under perturbation if the target has no route for correcting it when conditions change. Conversely, a newly calibrated instrument can be answerable before it has much of a track record: standards, logs, and downstream checks already specify how discrepancies would show up. Reliability is a statistical property of a process; answerability is a counterfactual property of a network. Corrective control is architectural in the same way; what a track record estimates is how effectively it gets realized.

A concrete transfer case shows the pressure. Suppose a retrieval-augmented system, which retrieves external records and places them in the model's context, scores well on source-backed factual questions about clinical guidelines. A coarse process description such as "this system answers medical questions reliably" invites transfer to neighbouring medical tasks. The profile view withholds that transfer. The success was supported by record access, text extraction, and institutional testimony. It should transfer to nearby record-backed questions and degrade on tasks whose missing stabilizer is live measurement, patient-specific observation, calculation, or intervention.

A careful reliabilist can individuate the process more narrowly, as retrieval-backed answering under a particular profile of records, tools, and checks. At that point the reliabilist has reproduced the profile-level carving. The difference is methodological: the profile account requires that carving before the transfer is granted.

Two cases show where the accounts come apart in practice. A newly calibrated instrument has no track record of its own, while its standards, logs, and downstream checks already fix how a discrepancy would show up. Run it the other way: a long-reliable procedure whose checks are quietly withdrawn keeps its record intact while the routes that produced it are gone.

Neither case refutes reliabilism. A reliabilist who assesses process types rather than individual track records handles both. The new instrument inherits the reliability of a type it belongs to by calibration, and withdrawing the checks changes which type is in play. The reliabilist can re-individuate after either case, and will.

What that concedes is smaller than it looks. Type membership has to be fixed by something, and re-individuating requires knowing which checks were carrying the reliability, which is what the profile supplies. So the disagreement isn't about entitlement to project. It's about which variable predicts when projection fails, and the profile names one: a reliability estimate carried across a change that removes the cross-source checks should degrade, and an account that leaves that variable out will predict worse. Section 7.7 says how to test it.

5.1 Adjudicating mechanism conflict

When two stabilizers disagree, say perception reports one thing and well-attested testimony another, a theory focused on one source at a time underdescribes the evidential situation. Goldman-style reliabilism can ask which process type is reliable, including conditional reliability; social reliabilism can extend the assessed process through testimonial production (Goldberg, 2010; Goldman, 1979). Those are genuine resources. The profile question is different: how tightly have these sources been calibrated against one another in this domain?

In Khalidi's terms, causal relations among associated properties underwrite projection; here the relevant projection runs from one stabilizer's output to another's (Khalidi, 2013).

Consider a modest conflict. A subject's visual judgement says that an object has property *P*. A calibrated instrument says that it lacks *P*. Several competent people, drawing on the same instrument tradition, say the instrument is functioning normally. The profile view avoids a standing rule to trust perception, testimony, or the instrument. It asks which sources have historically corrected which others in this region of practice.

In everyday object recognition, ordinary perception, practical feedback, memory, and testimony have long corrected one another. If I seem to see an old friend at the end of a hallway while well-placed testimony rules out her presence, the perceptual judgement can reasonably give way. Coady's discussion of perception, memory, inference, and testimony already pushes in this direction: the sources interpenetrate, and testimony can rationally undermine what presents itself as perception (Coady, 1992).

In a mature instrumental domain, the priority can reverse. Unaided perception may be the weak node, because the tighter profile joins calibrated instruments, records, expert testimony, replication, and repeatable intervention. The same abstract conflict receives different treatment in different domains. Global source reliability is too coarse a contrast. The history of cross-source correction gives one output a stronger projective relation to the rest of the profile.

Co-variation carries epistemic weight. A tightly coupled profile gives a defeasible presumption that a lone divergent output is local error or a boundary case. A loosely coupled profile gives a different presumption: divergence is evidence that the kind of success under assessment projects weakly here. The verdict is still empirical and fallible, but both the pattern of calibration among stabilizers and the assessed reliability of one source enter the evidence.

Agreement alone doesn't guarantee truth. It can be manufactured, coerced, or inherited from a shared mistake. Agreement produced by independent, mutually correcting routes has a different evidential meaning from agreement produced by one copied source. The profile account keeps that difference visible.

Russell's classic objection to the coherence theory of truth made the structural point early: a false proposition can cohere with some specified set, so coherence alone can't select truth (Russell, 1907). The profile account inherits the lesson without adopting Russell's theory. Coherence stabilizes as one route among several, and its force depends on its coupling to input and correction routes.

Semantic fragmentism gives the coherence-only case its most developed contemporary statement. Havlík (2024) argues that linguistic meanings are grounded in a corpus rather than in the world or in any internal representation of it, a corpus counting as minimal once further additions leave the grounding of its parts unchanged.

His target is meaning rather than truth, so Russell's objection doesn't transfer as a refutation. The structural parallel holds anyway. A saturated corpus fixes the meanings of its parts whether or not those parts are true of anything, and nothing inside it registers drift. Havlík grants that models are sensitive to their training data, and that errors in it degrade what they do. Errors enter by that route, and the account supplies none by which they leave.

Bayesian work on witness independence and coherence establishes the evidential point: agreement confirms more when sources are sufficiently independent than when they share a source or an error process (Bovens & Hartmann, 2003; Olsson, 2005). The profile account uses the point differently. Calibration history becomes an individuation principle for systems and transfer as well as evidence for a body of testimony.

Transfer cases make that difference matter. Suppose a model, instrument, or witness has been reliable in a domain where perception, records, and measurement have repeatedly corrected one another. A source-by-source assessment can preserve that reliability score when the source is moved to a neighbouring domain.

Here the profile view withholds that presumption. If the new domain lacks the calibration history that made the earlier success projectible, it predicts weaker transfer even before any present disagreement appears. The divergent prediction is that reliability carried apart from its cross-source correction background should degrade under perturbations that remove the missing checks.

5.2 Domain-clustered co-variation

A network view also foregrounds a cross-domain pattern that source-by-source accounts tend to leave implicit: disagreement should cluster where the independent marks of connectivity are thin. Khalidi plots degrees of projectibility along two dimensions, the generality of the projections a predicate supports and their variety, taking these to be orthogonal though each can compensate for weakness in the other (Khalidi, 2018). The stabilizers of truth-relevant success should be densely linked where they've been jointly disciplined, and sparsely linked where they haven't. Persistent disagreement should cluster in the sparsely connected regions.

To avoid circularity, connectivity has to be characterized independently of agreement. The graph still requires a purpose: Onishi and Serpico (2022) show the abstraction level and boundaries stay interest-relative.

Replying requires fixing the purpose while leaving the interest-relativity visible. Holding the representational aim of Section 4 fixed, connectivity can be characterized by three marks established before present agreement is checked.

First, the stabilizers can share a selection or training history. Perception, naming, action, and memory in ordinary object recognition have been shaped by recurring tasks and recurring error signals. Instruments, calibration routines, and expert reports in mature scientific practices can share a similar history, even when the relevant objects aren't available to unaided perception.

Second, the practice can have accumulated correction procedures. Longevity by itself isn't enough for reliability; a long-lived mistake remains a mistake. What matters is the accumulation of correction norms, records, role differentiation, and procedures for passing error reports forward. These features make one stabilizer available as a check on another before any present disagreement has been adjudicated.

Third, the domain can have measurement infrastructure. Instruments, standards, replications, and intervention routines let a practice ask whether perception, testimony, or coherence has drifted. Where those infrastructures are thick, the stabilizers can correct one another. Where they're thin, the same stabilizers may continue to operate, but their relations support weaker projection.

Used this way, the marks are defeasible indicators, not criteria. Disagreement can also be driven by stakes, incentives, ideology, small samples, strategic ambiguity, or value-ladenness. The profile view predicts especially persistent disagreement where those familiar pressures combine with weak cross-source correction.

Once the purpose is fixed, the profile predicts tighter agreement where shared training history, institutional correction, and measurement infrastructure are thick. It predicts looser agreement where those marks are absent. Frontier science and remote historical fact are expected edge cases for that reason: some stabilizers remain real while the links among them thin. Whether the pattern extends to normative disagreement is a further question, and Section 6 leaves it open.

Profile structure adds to the component theories by explaining changes in relative epistemic force. Process reliability, coherence, correspondence, pragmatic success, and testimony remain local successes. Their force changes across domains because the calibration background changes. Section 7 applies that result to language models: systems can inherit textual and coherentist stabilizers with attenuated perceptual and verificational links.

6 Objections

6.1 Interest-relativity

One objection cuts at the profile structure itself. Onishi and Serpico (2022) argue that Khalidi's causal-network account inherits the interest-relativity Craver (2009) finds in mechanistic accounts: which causal graph you draw, at what level of abstraction, and where you set its boundaries all answer to explanatory interest, so the network never fixes a unique, interest-free carving of kinds. If that's right, the profile of truth-relevant success is only as objective as our purposes.

The objection assumes that an account of kinds owes an interest-free carving. Khalidi's own doesn't. He treats the sciences as studying a jumble of partially overlapping domains, each with its own toolkit of categories and methods, and holds that where two schemes sort members into categories that don't coincide there's often no ground for ruling either scheme non-natural (Khalidi, 2013). Isotopic and isobaric nuclides carve the same atoms differently. Neither is the privileged carving, and the projections each supports hold up.

The reply has a metaontological cousin. On an easy approach to ontology, existence questions are settled by trivial inferences licensed by a concept's rules of use, which moves the substantive work from whether the entities exist to which concepts we should employ and why (Thomasson, 2015). I take no stand on that general thesis, and the account doesn't need it. What it borrows is the order of business. First ask which profile concept to adopt for which inferential and corrective purpose. Then ask the question the world settles, whether that concept's declared projections succeed. The second isn't answered by the first, which is what keeps the account from reducing to a report of our interests.

So purpose-relativity isn't a concession wrung from the account. It's the condition under which the account says anything at all. What matters is whether the purpose is declared before the verdicts are collected, and whether the claims it licenses can fail. Held to the representational purpose of Section 4, two claims follow: systems missing perceptual and verificational stabilizers should show the failure pattern of Section 7, and connectivity should track selection history, accumulated correction procedures, and measurement infrastructure (Section 5).

A sharper version of the objection would bite. If the purpose were chosen after the verdicts were in, to secure them, the account would be unfalsifiable. Section 5 blocks that by characterizing connectivity through marks established before present agreement is checked. Declaring the interest is what makes the account refutable, not what excuses it.

6.2 Relabeling and absorption

A second objection says that the account will be absorbed by the component theories. Reliabilism can already conditionalize process reliability. Social epistemology can extend the assessed process through testimony, records, and institutions. Coherentism can demand input constraints. On this objection, the profile view redescribes familiar epistemology in causal-network language.

Component theories often have resources to mention coupling. Still, profile structure makes the calibration history itself the unit from which we project. Section 5 gave the central case. A source can retain its local success record while losing the correction background that made that success travel. The profile view predicts weaker transfer in that case, because the missing cross-source links matter before a fresh reliability score has been established. A careful reliabilist can reconstruct the same grain after the fact; the divergence is predictive rather than deontic. The profile names the variable whose absence predicts degraded transfer, and an account that ignores it makes worse predictions. For complex systems, that move raises the generality problem for reliabilism at system scale (Conee & Feldman, 1998). Pretraining, filtering, RLHF, retrieval, tools, prompts, and institutional uptake can be carved into many process types. If the reliabilist uses selection history, accumulated correction procedures, measurement infrastructure, and architectural facts to choose the projectible carving, the profile view has supplied the individuation theory.

In this respect, the account complements reliabilism, coherentism, correspondence, pragmatism, and social epistemology. It says when their local stabilizers support projection together, when they pull apart, and what failures should follow.

6.3 Derivative testimony

A third objection concerns testimony. Testimony is an interpersonal epistemic practice involving assertion, responsibility, trust, uptake, and defeaters (Coady, 1992; Goldberg, 2010; Lackey, 2008). If so, giving a language model testimonial standing because it was trained on human text alone misleads. Recent work on artificial intelligence testimony presses the same role question from the other direction: conversational systems strain anthropocentric theories of testimony while still differing from ordinary human testifiers (Freiman, 2024).

Accepting that point gives the weaker claim. A text-trained model inherits artifacts of testimonial practices: reports, explanations, records, instructions, arguments, and corrections left in text. That inheritance is a genuine stabilizer because many such texts were produced inside practices already partly accommodated to the world.

Training alone also leaves the stabilizer derivative: it gives the model neither the role of a normal hearer who can trust a speaker under standing norms of assertion nor the role of a speaker who takes responsibility for what's asserted. That derivative status explains both the reach and the fragility of text-only systems.

Sharpened, the objection isn't yet answered by that concession. Causal descent from testimonial practice isn't participation in it, and something has to survive corpus selection, tokenization, optimization, and generation for epistemic support rather than mere influence to be inherited. So the claim needs a preservation condition, and the distributional character of the inheritance supplies one. What survives robustly is whatever shapes the distribution of forms: the shape of a practice, its characteristic moves, the regularities that hold across many reports. What survives only sparsely is what a particular passage carried rather than the pattern, so source identity, provenance, and defeaters are inherited less robustly and recovered less reliably, which is why verbatim anchors get fabricated at the rate they do. The contrast is one of robustness, not of presence and absence.

Call the surviving relation DOCUMENTARY INHERITANCE if "testimony" strains too far. The label matters less than the condition, which is checkable in the way Section 7 sets out: properties recoverable from distribution should be inherited, and properties carried by individual provenance shouldn't be.

6.4 Outputs as fiction

A fourth objection denies that the outputs are the right kind of thing to assess. Ruyant (2026) applies accounts of epistemic representation from philosophy of science and concludes that a language model represents a constructed class of appropriate linguistic production, fixed by its makers' norms, rather than the world. Conversation with a chatbot is then a fiction, and its outputs are "props in a game of make-believe". Auditing props for answerability would be a category error.

Two things about it are right. It reaches by a different route the same fictionalist conclusion others have argued for directly. And it states its own scope: Ruyant's generalization holds "to the extent that" such systems aren't trained on direct worldly inputs, which is a claim about routes in another vocabulary.

Ruyant supplies the reply himself. He grants that fictions can be performative, since generated text becomes student dissertations, professional emails, computer programs, news and scientific articles once a human appropriates it. His analysis stops there, and the ethical conclusion he draws next, that deployment promotes applications for which such systems are "fundamentally untrustworthy", needs an answerability notion his account doesn't supply.

Appropriation doesn't have to convert one kind of object into another, and the profile doesn't need it to. One string can be correct in the game, false about the world, effective as an artifact, and something a user is answerable for, all at once. Those are different correctness targets, and the account is target-indexed already.

So the question isn't whether the output is still a prop. It's which target is being assessed, and which routes were live for it: the ones active when the text was produced, and the ones the appropriating practice adds. A draft taken into a practice with measurement, review, and correction has a different profile against the world as target than the same draft pasted into a filing, while remaining the same string and, in the make-believe, the same prop. That extends the fictionalist analysis rather than refuting it.

6.5 Non-empirical domains

A fifth objection comes from mathematics, logic, ethics, fiction, and law. In these domains, perception, measurement, intervention, and practical success seem peripheral or misplaced. If the profile is built around world-contact, it may overfit ordinary empirical truth.

Narrow the scope and the objection loses its grip. This account targets world-directed representational success, where getting it right means getting some independently variable matter right, and where the LLM argument lives. Mathematics is instructive as a contrast rather than as an extension: proof and formal checking supply detection and correction of their own, so an arrangement can be under corrective control there with no perceptual or instrumental route at all. That the architecture has non-empirical instances is a point in its favour; it isn't a claim that this paper has characterized them.

Law, ethics, and fiction I mark as possible extensions and leave open. Institutional procedure, role differentiation, precedent, and contestation look like discrepancy handling in the sense used here, but saying so would need the kind of case work the empirical argument has received, and no later section draws on the comparison.

6.6 Deflationism and pluralism

Finally, deflationists and alethic pluralists may object from opposite directions. Deflationists may say that truth has no deep nature for the profile to explain (Horwich, 1998). Pluralists may say that different domains involve different ways of being true, so a single profile is artificial.

Both objections help fix the ambition. The paper explains projectible patterns among truth-relevant representational successes. That target is compatible with a thin account of the truth predicate. It also fits domain pluralism, because the profile already allows different stabilizers to be privileged in different regions. The claim concerns the projectible stabilizer profile of truth-relevant success.

7 The LLM case: participation, not a verdict

7.1 The bearer and the modes of participation

Start with the bearer, because the obvious candidates don't work. A profile doesn't belong to a model, since the same model has different profiles on different tasks. It doesn't belong to an output, which has no routes of its own. And it can't belong to the whole surrounding practice, or every deployment inherits the epistemic credit of human science.

The bearer is the ARRANGEMENT: whatever produces the output, indexed to a target, a task, and the routes live for that task. An arrangement may include the model, the prompt, retrieved records, tools, sensors, checkers, human reviewers, and update channels. Fixing it this way settles several questions at once. Retrieval belongs to the arrangement when it runs, not to the model. Human appropriation changes the arrangement rather than adding a gloss to a fixed one. Inherited participation belongs to the arrangement's causal history. And a nominally multimodal or tool-using system reverts to a text-only profile whenever those channels sit idle, which is why the same product earns different verdicts on different work.

Different arrangements then have different profiles. A pretrained text model, an instruction-tuned chat model, a retrieval-augmented system, a tool-using agent, a multimodal model, and an experimentally embedded robot have different stabilizing backgrounds. The theory asks us to compare them: change the stabilizers, and the expected successes and failures should change too.

Participation names the key relation. An arrangement participates in a stabilizer to the extent that its outputs are generated, constrained, corrected, or selected by that stabilizer in a way that supports projection to further cases.

Participation can be inherited, as when training text contains the residue of human perception, testimony, measurement, and correction. It can be operational, as when retrieval, tools, sensors, or instruments constrain the present output. It can be corrective, as when feedback changes later behaviour. It can also be decorative, as when an output mentions experiments, sources, or measurements with the relevant constraint absent.

These are modes, not bins. A retrieval that returns a passage half-containing the answer constrains the output partly, and decorative participation is the limit of operational participation as the constraint goes to zero. The name earns its place because that limit is common and has a characteristic look, which is the language of measurement without measurement.

Corrective participation divides by horizon, and in-context correction is the case that makes the division visible. A user who supplies a counterexample mid-session can change what the arrangement produces from that turn on, which satisfies effective uptake within the session and nothing beyond it. The weights don't move, the next session starts where the last one began, and no other user benefits. That's real corrective participation with a horizon of one conversation, and distinguishing it from correction that reaches policy matters for the same reason the inherited case did: what looks like one route is two, differing in how far the repair travels.

The four modes describe how arrangements instantiate the features of Section 4, often several at once rather than one apiece. Section 3 said a single route can occupy several positions, and the same holds here: a live measurement can supply target access, register the discrepancy, attribute it, and drive the revision. What follows are characteristic contributions, not assignments.

Inherited participation supplies DERIVATIVE ANSWERABILITY, and supplies it only on a condition. Discrepancies already registered in the source practices shape present production through the training history, which is a delayed route of the kind Section 2 allows. Causal descent isn't enough by itself. Dense propaganda leaves distributional traces as robust as a measurement literature's, and a system trained on it inherits the shape of a practice with no truth-sensitive signal in it. What has to survive the training history is the residue of corrections made against the world rather than against an audience, a rater, or a party line.

Three things come apart here. Causal inheritance, which any dense corpus supplies. Derivative answerability, which needs the inherited corrections to have been world-directed. And live answerability, which needs a route from a fresh discrepancy to the current output. Figure 1 sets the three out.

Inherited participation never supplies live answerability. Once the training history is fixed, a discrepancy arising after it has no way to reach production.

Training against verifiable rewards tests this. When a model is optimized against unit tests, a proof checker, or execution results, a discrepancy between output and an independently specified target is detected and does alter production. All five features can be instantiated there, to the extent

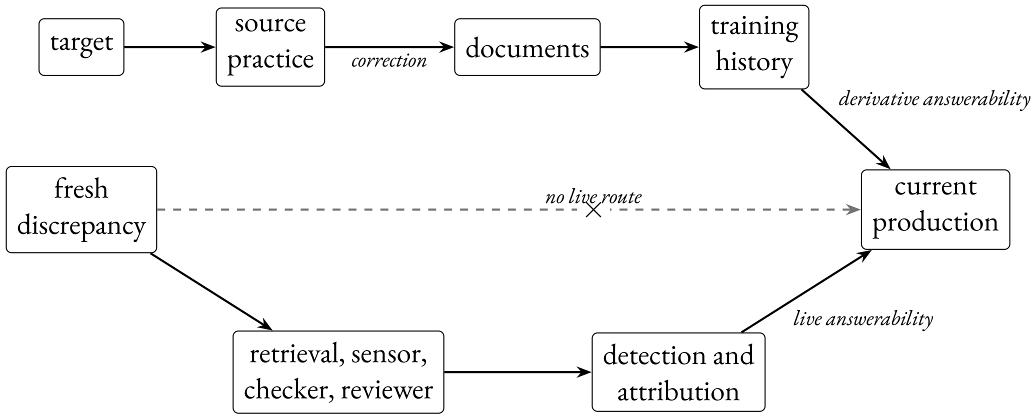

Figure 1: Three routes to current production. The historical route supplies derivative answerability, and only where target-sensitive correction survives in the training distribution: dense causal ancestry on its own doesn't, which is why a propaganda corpus leaves equally robust traces and no answerability. A discrepancy arising after the training history is fixed has no route of its own. It reaches production only where a live route is active in the arrangement, which is a fact about the arrangement rather than about whether the weights can change.

that the verifier is adequate to the target, sufficiently independent of the generator, and attributable in its signal. Nothing guarantees that, since tests are incomplete, attribution is often coarse, and a verifier's errors can correlate with the generator's. Where it holds, it holds at training time.

A model trained that way has been corrected; it isn't thereby correctible. Verifiable-reward training delivers corrective control over the targets its verifier covers, while it runs. It delivers none over targets the verifier doesn't cover, and none at inference, where no verifier is present and the weights are fixed. A model that has learned arithmetic against a checker still can't tell you whether today's figure is right. Verifiable rewards make the derivative answerability sharper, since the corrections they encode were made against an independently specified target rather than against a rater. They don't make it live.

Operational participation characteristically supplies target access. Corrective participation is where detectability, attribution, and effective uptake usually enter, and only when the feedback is attributable and reaches production. Decorative participation supplies none of them while resembling all. Checking-route independence belongs to none of the modes in particular, since it depends on where a checker came from and what it can see rather than on how it participates.

Corrective control is bounded by the union of the arrangement's live routes, and it's bounded target by target. A user who checks a claim adds corrective control for that claim at the rate they check, and none for the next one. A checking step that's available and routinely bypassed adds the look of one. So the interesting case isn't an arrangement with no corrective control anywhere, which is rare, but the ordinary one in which a system is asked about targets that no participant can independently check. Grounding is untouched by that boundary. Corrective control stops at it.

With these distinctions, participation doesn't collapse into metaphor. Inherited participation explains why a text-only system can know its way around a domain whose records are dense and stable. Operational participation explains why retrieval and tools can matter at deployment as well as during training. Corrective participation explains why feedback matters only when it changes future behaviour in ways tied to the relevant error. Decorative association is the failure mode: the language of measurement without measurement, the language of citation without a record.

They also block a circular reading of the account. Causal and architectural facts individuate participation before the performance test: training objective, data provenance, filtering history, runtime information flow, tool-call logs, sensor access, update channels, and the route by which feedback changes later behaviour. Performance then tests whether those architectural facts were the right stabilizers to mark.

7.2 Route-specific interventions

The main interventions can be described in stabilizer terms. Retrieval augmentation queries an external corpus or search system and places retrieved material into the model's context; it adds record access when it works. Its characteristic failures are diagnostic rather than incidental: a stale index supplies a record of the wrong time, a chunk boundary supplies half a record, and an embedding match on wording rather than content supplies a passage that resembles the answer. Each leaves the archival route nominally present and materially absent, which is decorative participation arriving through a mechanism rather than through a turn of phrase. Tool use routes a subtask through a calculator, database, code interpreter, application programming interface, or instrument; it adds calculation or measurement only when the right tool is called and its result is integrated. Coding adds a stronger correction loop when generated code is run: a compiler error, traceback, failing assertion, or passing test supplies fast, attributable feedback about the produced artifact.

Self-consistency prompting samples multiple completions or reasoning paths and privileges convergence among them; it strengthens internal convergence. An external check requires a separate route to the relevant facts. Instruction tuning and RLHF change the model's policy through demonstrations, preferences, benchmarks, or evaluations; they add a social or evaluative correction channel whose target depends on the feedback.[4] Multimodal input supplies image, audio, video, or sensor information; by itself, it adds perceptual input. Calibrated measurement is a separate stabilizer.

Multimodality matters because it seems to answer Harnad's sensorimotor prescription. The profile gives a finer-grained diagnosis. Harnad's proposal joins perception to action, correction, and use (Harnad, 1990). A vision-language model trained on image-caption corpora inherits perceptual participation through selected, framed, and captioned images. A deployment-time image input adds operational perception. Those routes supply perceptual input; calibrated measurement, intervention, and action-guided correction remain separate stabilizers.

Multimodal input should improve performance unevenly. It should reduce some failures in object classification and leave measurement- and intervention-dependent tasks fragile. It should also leave room for object hallucination, where generated descriptions include objects not present in the image (Y. Li et al., 2023). That pattern is what the profile view expects when target access is strengthened and the correction routes are left as they were.

To test the view, hold the topic, answer format, and apparent difficulty fixed, and vary the missing stabilizer. Gains should follow the added routes. Retrieval should improve source-backed recall

[4]Preference signals have factual targets. Rater approval of affirmation, deference, or confident presentation is a fact about a population and interaction setting, and RLHF can make the model's behaviour strongly answerable to that fact. The failure mode is target mismatch: the policy can track the rater distribution accurately while its assertions are assessed against a different, object-level target. Sycophancy is the clean case, where a system accurately tracks the wrong target.

more than novel measurement or action-sensitive causal judgement. Self-consistency should improve coherence-sensitive reasoning more than claims that need a fresh route to the facts. Multimodal input should improve perceptual classification before it improves calibrated measurement. If gains transfer evenly across those contrasts, that would be evidence that the profile has carved the stabilizers badly.

A deployed product may occupy several rows at once. A chat system can be instruction-tuned, retrieval-augmented, partly tool-using, and multimodal. The profile question asks which stabilizer is live in the task being assessed.

Which profile counts as the baseline depends on the task. A deployed system with retrieval, tools, or multimodal channels occupies the baseline whenever those channels sit idle or appear decoratively. The text-only case is the profile to which richer systems revert when added stabilizers are unavailable, inappropriate, or unintegrated.

As a pressure test, the text-only baseline gives the cleanest case. It's a limiting, task-relative position in the profile space rather than a description of a product. Systems trained on interleaved image and text, on code corpora, and against verifiers occupy richer positions, so a purely text-trained model is a reconstruction. It earns its place because richer arrangements revert to it for any task whose added routes are idle, inappropriate, or unintegrated.

Such a model inherits textual products of human accommodation: reports written by perceivers, explanations corrected by teachers, records produced by institutions, descriptions shaped by instruments, and genres whose norms often reward coherence and accuracy. That inheritance is why text-only systems can support genuine epistemic achievements. It also explains why the inherited stabilizer deserves derivative testimonial status. The model has access to testimonial residue, a different role from that of a speaker or hearer in a testimonial exchange.

Table 2 summarizes the architecture-sensitive predictions. The rows are profiles rather than a scale from bad to good.

System profile	Added or dominant stabilizer	Directional prediction
Text-only pretrained model	Textual inheritance and local coherence	Strong where dense, stable text is a good proxy; fragile for recent, obscure, perceptual, or measurement-dependent facts.
Instruction-tuned or RLHF model	Preference, helpfulness, safety, and benchmark feedback	Better conversational fit and some error reduction; independent routes to the facts still have to be supplied.
Retrieval-augmented system	Live access to records and institutional traces	Better on updated or source-backed recall; remaining fragility where the missing route is experiment, perception, or intervention.
Tool-using or coding system	Calculators, databases, code execution, instruments, or application programming interfaces	Better where the right tool is called and interpreted; strongest when runtime or test results guide revision; new failures from tool choice, stale data, and test-as-proxy effects.
Multimodal system	Perceptual input in image, audio, video, or sensor form	Better for some perceptually anchored tasks; calibrated measurement and practical correction remain separate stabilizers.
Action-guided or expert-reviewed system	Practical feedback, review, and downstream failure pressure	Stronger correction profile where feedback is timely and domain-competent; weaker where review is sparse or proxy-based.

Table 2: Architecture-sensitive profile predictions.

7.3 Inherited structure and missing anchors

Derivative inheritance cuts both ways. Human text is saturated with world-directed practice. It contains the residue of people looking, measuring, arguing, repairing, citing, and correcting one another. The model usually remains outside those practices when it produces a fresh answer. It draws on their products with attenuated access to their present checks.

Text-only training can still produce more structure than a bare "textual residue" slogan suggests. The generative pretrained transformer model Othello-GPT is an important pressure case: a sequence model trained to predict legal moves in a board game developed an internal representation of board state, and interventions on that representation affected its outputs (K. Li et al., 2023). Such cases fit the profile view where the target structure is densely and systematically encoded in the sequence stream. They show that inherited participation can be strong when the relevant world is small, rule-governed, and recoverable from the training signal.

Othello is a thin base for a claim this size. It's finite, fully observable, and its legal moves are a deterministic function of a state the move sequence determines, so the training signal contains the world entirely. The result is striking for that reason and unrepresentative for the same one. Nothing follows about domains where the state isn't recoverable from the record, which is the class the argument concerns. What the example establishes here is narrower, that inherited participation isn't limited to surface statistics when the world is small enough to fit in the stream.

Limits matter. Internal world-model evidence should be strongest where the relevant state variables are encoded in the sequence and the loss function penalizes violations of their structure. It should weaken when success depends on variables absent from the sequence, live measurement, changing local conditions, or intervention in the target system. Robust transfer across those boundaries would count against the profile diagnosis.

After inherited testimony, coherence is the second stabilizer these systems participate in strongly. Training and decoding reward genre fit, local well-formedness, smooth inference, and continuation from context. Those pressures explain why these systems can display impressive formal linguistic competence, in the sense separated from functional world use by Mahowald et al. (2024). They also explain the vulnerability. Coherence can improve the shape of an answer and add no independent route by which the relevant facts constrain it.

In the baseline profile, perception, measurement, intervention, and practical correction enter weakly through text produced by others. A model can generate a plausible lab result, legal summary, local observation, or bibliographic claim because similar text exists. The present output remains unconstrained by an experiment, court record, observation, or catalogue entry.

Textual inheritance is reliably distributional and only contingently archival. Training can and does preserve particular strings, names, and sources, so the claim isn't that token-bound information never survives. It's that nothing preserves it under a general relation of record identity, provenance, and current checkability, so what survives robustly is what's redundantly represented across contexts and what survives sparsely is what a single passage carried. A text-only model inherits statistical regularities in testimonial practice, so it can navigate a domain whose records are dense and stable while fabricating the verbatim anchors (quotations, identifiers, author lists) on which the practice's correction routines depend. The same distinction predicts that these failures should be especially retrieval-sensitive: retrieval adds the archival route that distributional inheritance lacks.

Hallucination plays a more specific role here. The LLM literature treats it as heterogeneous (Huang et al., 2025). The profile claim is narrower: text-only systems should be especially vulnerable to fluent, coherent, genre-appropriate outputs that fail when the task requires a fresh or independent route to the facts. Those failures should be common for obscure sources, recent events, local observations, novel measurements, fine-grained quantities, and action-sensitive causal claims. They should be less common where dense, stable, redundant text is already a good proxy.

Frankfurtian accounts of bullshit (Frankfurt, 1986) diagnose LLM output differently. Hicks et al. (2024) apply that diagnosis to ChatGPT-style systems, arguing that their outputs are produced with indifference to truth. On the Frankfurtian diagnosis, the system doesn't care where the facts are; on the profile diagnosis, the relevant facts have no route into production or correction. That missing route yields comparative predictions across retrieval, tools, multimodality, and action-guided feedback.

Mitigation results should be read through the same distinction. Retrieval can reduce hallucination in conversation (Shuster et al., 2021); that's evidence about record access. Perception, experiment, and intervention require further routes. Tool use and multimodality make parallel changes. They add operational routes where the tool, sensor, or modality is live in the task; otherwise the new vocabulary of tools or perception can remain decorative.

Action-guided systems provide the strongest possible upgrade in the table when the feedback is timely, attributable, and tied to the relevant failure. Coding is the clean artificial case. A system that runs generated code, observes a compiler error, traceback, failing test, or successful execution, and revises against that result has a tighter corrective route than a system rewarded for plausible-looking code. A robot that breaks a glass, misses a target, or fails a laboratory protocol can receive similar correction unavailable to a text-only model. A deployed chatbot that receives delayed approval, engagement metrics, or sparse user edits has a thinner corrective route. The label "feedback" hides that difference unless the stabilizer is specified.

7.4 What the correction channels correct

Instruction tuning, RLHF, benchmark evaluation, red-teaming, and expert review function as correction channels, and they should be classified by what they correct. They often impose helpfulness, harmlessness, preference satisfaction, conversational fit, and benchmark success. Those constraints can reduce some errors and can also reward answer-shaped behaviour when users or raters prefer confidence, agreement, or plausible explanation.

Post-training isn't one thing, and the profile splits it in two. Where the reward comes from a rater's approval, the target is the rater. Where it comes from a unit test, a proof checker, or an executed result, the target is whatever the checker checks, and the correction is world-directed in the sense that matters here. Both are called post-training and they build different profiles. Treating the second as preference feedback understates what it supplies; treating the first as truth feedback overstates it. The relevant question is never how much post-training an arrangement received but which targets its rewards were computed against.

Work on sycophancy shows the risk of treating preference feedback as truth feedback on its own (Shapira et al., 2026). In the profile terms, these processes strengthen some social and evaluative stabilizers and leave perception, measurement, and intervention proxy-based.

What fine-tuning does is disputed, and the disagreement dissolves rather than needing a side. Grindrod (2024) treats it as a relatively marginal procedure that can't be what makes the difference between intentionality and its absence, since models produce apparently meaningful text without it. Ruyant (2026) treats it as constitutive of what the system represents, because the norms of appropriateness a model is tuned toward are what fix its target.

Both are right about different stabilizers. Fine-tuning is constitutive of appropriateness for a purpose, which is Ruyant's target and is fixed by whoever sets the norms. It's marginal to world-directed correction, which is Grindrod's, because preference feedback supplies that only when the feedback is itself generated through a target-sensitive route: verified factuality, tool results, or action outcomes rather than approval. Reading the disagreement as one about a single quantity called grounding is what makes it look unresolvable.

7.5 What grounding leaves open

Coelho Mollo and Millière press the strongest neighbouring challenge. They distinguish kinds of grounding and argue that some LLM states can satisfy causal-historical or teleosemantic conditions for referential grounding (Coelho Mollo & Millière, 2026). Their conclusion isn't in dispute. The profile view's explanandum is broader than referential grounding.

Work on neural word embeddings pushes in a similar direction: learned representations may have genuine, though limited, contents fixed by the functions they serve inside the system (Mallory, 2026).

Teleosemantic pressure of this kind fits the baseline text-only case. The reply concedes content and asks which stabilizers that content participates in.

Section 1 listed the routes to scepticism that recent work has closed: natural histories securing reference, intentions turning out to be the wrong place to look, the argument from communicative intention failing at both premises, and linguistic metasemantics delivering meaning where mental metasemantics doesn't. One addition belongs here rather than there. Pepp's positive suggestion is that the live question concerns a basic reference prior to propositional attitudes (Pepp, 2025), which locates the remaining work on the content side and so sharpens rather than blunts the point below.

Training text carries correction procedures, defeaters, and records, and a system can inherit some capacity to reproduce them. What inheritance alone doesn't deliver is a live route by which a fresh discrepancy reaches production and changes it.

Referential grounding covers one dimension of world-connection. Truth-tracking answerability depends on a wider profile of stabilizers, including correction, measurement, intervention, perception, and practical feedback. An LLM state might be referentially grounded in their sense even as the deployed system has a fragile truth-tracking profile for tasks requiring independent checking.

Mandelkern and Linzen make the gap vivid without meaning to. They close with Izzy, isolated from birth in a sensory-deprivation chamber and reaching the world only through a text screen, and they argue that externalism dissolves the puzzle about how his words refer (Mandelkern & Linzen, 2024). Izzy's answerability is nonetheless confined to the testimonial and inferential routes his screen supplies, with perception, measurement, and intervention absent. Secure reference alongside a radically restricted profile is the dissociation this paper is about, drawn by philosophers arguing the other side.

Pepp presses the harder version. She suggests that such systems may contact objects through their exchanges with users, coming into contact with whatever shaped a user's input (Pepp, 2025). The participation distinction splits that case. An uploaded image is an operational perceptual route, and the profile says so. Contact with the objects that shaped a user's text is inherited: the perceiving was done by someone else, and nothing in the exchange reports back when the user was wrong.

7.6 Corrective control in deployment

A deployed case shows what adding a checker actually takes. Anthropic's coding agent routes each proposed action through a classifier that blocks operations it judges irreversible, destructive, or aimed outside the working environment (Anthropic, 2026). The case concerns safety control rather than factual truth directly. Its role here is to isolate the architecture by which an independent discrepancy signal becomes behaviourally effective.

Three things follow, and they're the features of Section 4 in miniature. A separate classifier can supply detection and effective gating where the acting model registers nothing itself, which is detectability and effective uptake supplied by a second party rather than by introspection. Being a separate party doesn't by itself secure checking-route independence: an external checker can reproduce the generator's errors, and the same model given genuinely independent environmental information may not. Independence has more than one dimension, among them development provenance, the environmental information available to the checker, objectives, and conditional error structure. The reported hardening came from giving the classifier more of the second, including repository visibility and git state. Whether that also decorrelated its errors wasn't measured, and lineage alone wouldn't settle it, since added environmental information is the kind of change that can alter conditional error structure. Measuring it takes conditional error rates for checker and generator on a shared held-out set. And an approval step's contribution is whatever its exercise makes it, whatever its nominal standing: users reject 3% of individual permission requests against 39% of plans put to them for approval. The first isn't decorative outright, since it corrects in the cases where it's used. It's a route with low realized effectiveness, and decorative in the bypassed and rubber-stamped cases.[5]

7.7 Empirical transfer

Route-specific transfer gives the strongest test. Improvements in one route should remain limited where a task depends on a different stabilizer. If reliable perception, measurement, retrieval, tool use, expert correction, or action feedback fails to alter the expected contrast in the direction just specified, the profile has misdescribed the mechanisms. If text-only systems remain mainly dependent on inherited textual accommodation and local coherence, the same vulnerability should persist even as fluency improves.

A clinical test makes this concrete.[6] One item set asks for recommendations recoverable from guidelines or institutional records. A paired set uses the same disease area and answer format and requires a patient-specific lab value, observed sign, dose calculation, or intervention-sensitive causal judgement. The profile predicts a retrieval gain on the record-recoverable set and little transfer to the patient-specific set unless the needed measurement or tool route is supplied. Retrieval alone producing the same gain on both sets would count against the carving.

Matching the sets on difficulty cues won't do. An intervention that supplies a missing lab value also makes the item easier, so a difficulty explanation and a route explanation both predict a gain, and a cue-matched design can't separate them. Equating on measured baseline accuracy weakens the simplest version of the difficulty story without eliminating it: two sets can share a baseline mean and differ in discrimination, ceiling structure, response variance, latent subskills, or treatment-effect heterogeneity, and matching on noisy estimates invites regression to the mean. Treat baseline performance as a covariate and a design diagnostic, not as proof that difficulty has been removed.

The design that carries the weight is crossed rather than matched. Build item families in which the relevant route can be supplied or withheld within closely related items, cross route availability against task requirement, and estimate the interaction in a hierarchical model with varying effects for items, domains, prompt templates, and systems. Declare in advance the smallest interaction that would matter, report the interval rather than a verdict, and run the reasonable alternatives for item matching, scoring, and exclusion instead of resting on one sample.

That instruction should be followed here rather than passed on, so one declaration, for the retrieval case. The source feature is whether a live retrieval route over a fixed corpus is available. The outcome is substantive correctness of the recommendation, scored blind, which applies to both halves of the item set. Whether verbatim anchors resolve is a secondary outcome and applies only within the

[5] Same users, same system, two approval dialogues, which is consistent with route-specific non-exercise though it doesn't isolate the route from interface burden, action granularity, or perceived stakes. A controlled study points the same way, with human testers catching 13.6% of planted dangerous commands against the classifier's 89%, though its participants worked in a purpose-built environment and knew they were being evaluated (Anthropic, 2026). Both figures are a vendor's self-report about its own product and haven't been independently audited, which is worth saying in a paper that asks checking routes to be independent of the production route's error. They're used here to illustrate the availability and exercise gap, not to measure it.

[6] The design parallels validity argumentation in measurement: inferences from performance are licensed relative to a purpose and tested by varying method while holding the target construct fixed (Kane, 2013; Messick, 1989).

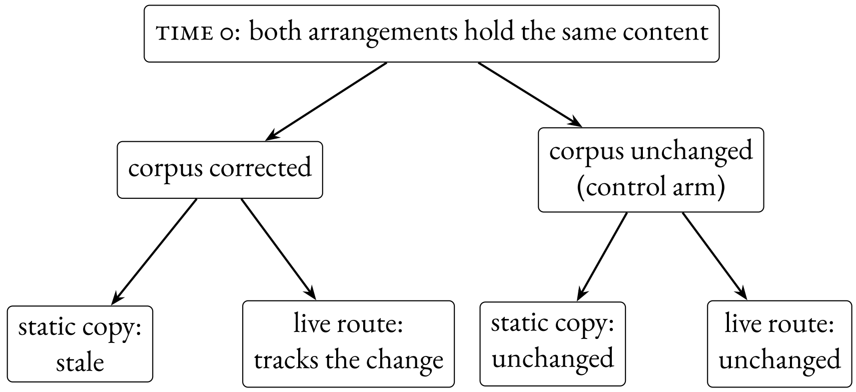

Figure 2: The discriminating contrast, with the profile's predictions shown. Substantive content is equated at time 0 by construction, so a coverage account defined over the information available to the arrangement predicts no difference between the two arrangements in either arm. The profile predicts a difference in the update arm and none in the control. No outcome magnitudes are implied, and none should be read in.

record-dependent half. Making anchor resolution primary would manufacture the interaction, since patient-specific items have no corpus anchors to resolve and would fail by construction rather than by hypothesis. The bearer is one named chat arrangement, retrieval switched on and off with the prompt template held constant, so the estimate is conditional on that arrangement and settles nothing about variation across systems. The population is a clinical item set split into record-recoverable and patient-specific halves, classified from the item text by raters who see no system output, and frozen before the first run.

Predicted is an interaction, with the retrieval gain on record-recoverable items exceeding the gain on patient-specific items by at least ten percentage points. Ten is a worked value rather than a derived one. Fixing it properly needs a decision basis the carving doesn't supply: the cost of a wrong recommendation, the cost of retrieval latency, and what a deployment would do differently at nine points rather than eleven. Stating it anyway is what makes the prediction refusable.

Interpretation then runs against ten rather than against zero. An interval concentrated above ten supports the claim. An interval concentrated near zero, or on the wrong side, counts against it. An interval running from below zero to well above ten reports an uninformative experiment rather than evidence either way, which is the case a bare significance rule would misread as a refutation. The classification isn't revised once the effect is visible. If it has to be, that's the result.

One interaction is weaker than two. The profile predicts that retrieval helps the record-dependent items and that a measurement tool helps the measurement-dependent ones, so crossing both interventions against both task types tests a double dissociation. A rival holding that useful information helps the tasks that need it predicts the main effects and not the crossing.

Accuracy alone still won't discriminate the profile from a coverage rival, on which retrieval helps wherever the answer happens to sit in the retrieved text. Scoring outputs twice, for substantive aptness and for whether verbatim anchors resolve, isn't sufficient either: a coverage account can predict that anchors improve more than aptness, since anchors can be copied from the retrieved material while aptness may already be near ceiling.

Contrasting a live tool against a pasted string doesn't settle it either. A capable coverage account can define coverage over the information available to the arrangement rather than over tokens already in the prompt, and a working lookup tool then counts as raising coverage dynamically. Any manipulation of how much can be got at is absorbable that way.

What isn't absorbable is a change in the world after the information is fixed. Give two arrangements the same content at the outset, one holding a frozen pasted record and one holding a live route to the corpus that record came from. Then correct the corpus, and ask a follow-up whose answer turns on the correction. Run a no-update control alongside it, in which the corpus is left alone, so the comparison isn't simply a live tool against a static context. Coverage is equated at the start by construction, so a coverage account predicts no difference between the arrangements in either arm. The profile predicts that the live route tracks the change in the update arm and matches the frozen string in the control.

Figure 2 sets the arms out. The contrast tests the paper's own property rather than a proxy for it, namely whether a discrepancy arising after production is fixed can reach the output.

It's also the manipulation a coverage account has the hardest time absorbing, because the two arrangements differ in no quantity of information at the moment they're equipped.

Those defeat a carving. What would defeat the framework is different in kind, and the difference is worth being careful about. Observing that every grounded arrangement in some sample also corrects well wouldn't show that grounding entails corrective control: both might depend on a third architectural feature, or every system sampled might satisfy more than the grounding conditions require. Non-entailment isn't refuted by co-occurrence.

What the framework does stake is explanatory. The five features are offered as the variables that account for which arrangements catch which errors and where improvement transfers. That claim loses if the features are manipulated successfully and repeatedly fail to explain the contrasts, or if some other variable explains them better with the features held fixed. Section 7.6 reports a case where a classifier improved through more environmental access with its error correlation unmeasured, which is one observation about a single feature, and one observation isn't a result.

8 Conclusion

Settle whether the representations inside a large language model connect to the world, and the question this paper asks remains open. An arrangement can acquire content, reference, and correctness conditions without acquiring the live, sufficiently independent routes by which a discrepancy with the target is detected and made to alter what it produces. Grounding doesn't entail corrective control.

Language models make the dissociation visible because they instantiate it at scale. They inherit dense textual residues of testimony and coherence, which leaves them derivatively answerable to corrections that have already happened, and they participate weakly in the routes that would detect a fresh error and carry it back into production. The profile says which routes an arrangement has, at what strength, for which targets, and so says when the gap has actually been closed rather than merely narrowed.

A thin truth predicate can coexist with thick truth-tracking practices. Where the links are actively maintained, the coupling can become homeostatic in Boyd's strong sense; elsewhere the profile can remain projectible below that threshold.

The older epistemologies keep their subject matter. Each can represent relations among the processes it describes, and reliabilists and social epistemologists routinely do. What the profile changes is the default unit: calibration history between sources becomes what we project from, rather than background against which a single process is assessed. The pattern can carry the epistemic fact.

Nothing here displaces those theories; their stabilizers interact. In ordinary cases, the interaction is easy to miss because the routes run together. In language models, the routes come apart.

A text-only model can inherit substantial textual residues of testimony and local patterns of inferential and discursive coherence. It may even acquire rich internal structure where the target domain is densely encoded in the sequence stream. That still differs from ordinary perceptual contact, calibrated measurement, intervention, and correction through action.

Comparison rather than classification follows. Retrieval, multimodal input, tool use, expert review, experimental embedding, and action-guided feedback alter the stabilizer profile. The question is which route they add, whether that route is live in the task being assessed, and where improvement should transfer. The prediction is an interaction between intervention and task, not a general gain: an intervention should help where it supplies the route the task requires and not otherwise. A system can improve along one dimension and remain fragile along another.

Empirical constraints remain. Participation is fixed by causal and architectural facts before performance is checked, and the predicted contrasts can fail.

Beyond artificial systems, LLMs serve as useful pressure cases because they separate mechanisms that human epistemic life usually runs together. They show that truth-tracking extends beyond linguistic competence, internal coherence, and statistical sensitivity to past assertion. Truth-tracking depends on a wider ecology of correction. Once that ecology is represented as a profile, the interesting question becomes visible: what a system participates in, what it borrows, what it lacks, and where those absences should matter.

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